3

CPU

Futex

3.1

StarryOS `axtask` `axsched`
Future Waker

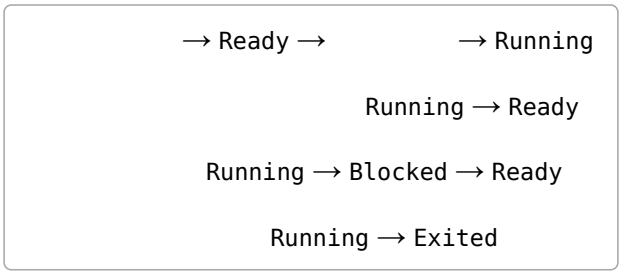
moss

`waitid`
CPU moss
Linux PollSet
Futex

3.2

3.2.1

- `Running` CPU
- `Ready`
- `Blocked`
- `Exited`



3-1

`Ready` `Running` CPU
`Ready` `Blocked`

CPU CPU
CPU

3.2.2

RR
5 CPU

`scheduler_timer_tick`



CPU idle

3.2.3

	`Running → Ready`	
`yield_now()`	`Running → Ready`	CPU
	`Running → Blocked`	

3-1

`switch_to`

`TaskContext` CPU

StarryOS

RR PID Linux

3.2.4 CPU

StarryOS CPU moss

`sched_getaffinity` CPU `sched_setaffinity` CPU

`set_current_affinity` CPU

`sched_yield` `sched_getscheduler`

RR moss

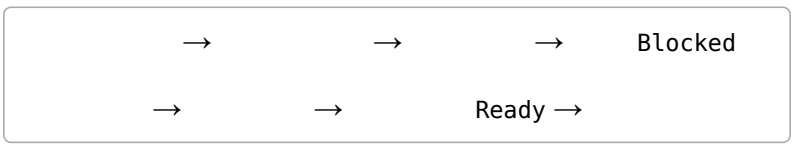
I/O CPU

3.3

3.3.1

`yield_now()`

`Blocked`



3-2

“ ”

3.3.2 Waker PollSet

Waker

Waker `wake()` Waker `Blocked` `Ready`

`PollSet` Waker eventfd timerfd pidfd

moss `has_waiters`

Waker

PollSet “ ”

Futex

3.3.3

`block_on` Future Future `Pending` `block_on`

`Blocked` Waker `Ready` CPU

Future

Waker `woke`

CPU

Waker

`interruptible` Future

`EINTR`

3.3.4

Waker

Waker moss PollSet

Futex 3.5

3.4

3.4.1

“ ”

`clock_nanosleep` Futex

3.4.2

moss `(deadline, seq)` `BTreeMap` `deadline` `seq`

alarm task

interval timer `TimerKey` Key

Key

`EVENT_NEW_TIMER` alarm task

3.4.3

`nanosleep` `clock_nanosleep` `TIMER_ABSTIME` moss

`EINTR` `rem`

LoongArch64

`yield_now()`

`Blocked` Futex

`timeout` Future

3.4.4

`TimeManager`

`times`

interval timer `ITIMER_REAL` `SIGALRM`
`ITIMER_VIRTUAL` `SIGVTALRM` `ITIMER_PROF`
`SIGPROF` interval

3.5 Futex

3.5.1

Futex

`FutexKey` Futex

Futex

Futex `owner_dead` moss

`notified` `bitset` `PollSet`

3.5.2 WAIT WAKE

`FUTEX_WAIT`

`EAGAIN`

Future `FUTEX_WAKE`

`notified` PollSet

wake

Waker `notified`

moss Waker

3.5.3

`FUTEX_WAIT_BITSET` `FUTEX_WAKE_BITSET` bitset mask

Futex

FUTEX_REQUEUE Futex
FUTEX_CMP_REQUEUE EAGAIN
notified bitset PollSet Waker
pthread

3.5.4

Futex moss
interruptible(timeout(...)) ETIMEDOUT EINTR
notified
FUTEX_WAIT FUTEX_WAIT_BITSET moss Linux
FUTEX_CLOCK_REALTIME

LoongArch64 futex_wait pthread cancellation

3.5.5 robust list

set_robust_list robust list Futex
moss futex_offset Futex owner_dead
EOWNERDEAD
ROBUST_LIST_LIMIT
clear_child_tid Futex
pthread_join

3.6

3.6.1

StarryOS
waitpid moss ProcessData
SIGCONT

3.6.2

wait
Waker child_exit_event
wake()
3.3

interruptible EINTR

3.6.3 wait

`waitpid` pid moss

`Pid` `Pgid` `Any`

`ECHILD`

`WNOHANG`	
`WUNTRACED`	
`WCONTINUED`	
`WNOWAIT`	

3-2

`wait4` `waitid` `siginfo_t`

`CLD_EXITED` `CLD_KILLED` `CLD_STOPPED` `CLD_CONTINUED`

3.6.4 waitid pidfd

moss `waitid` `P_ALL` `P_PID` `P_PGID` `P_PIDFD` PID

`P_PIDFD`

pidfd StarryOS moss `/proc` fdinfo PID

pidfd pidfd `EAGAIN`

pidfd

`WNOWAIT`

3.7

StarryOS RR moss

PollSet

`TimerKey`

moss Futex robust list

`waitid` pidfd